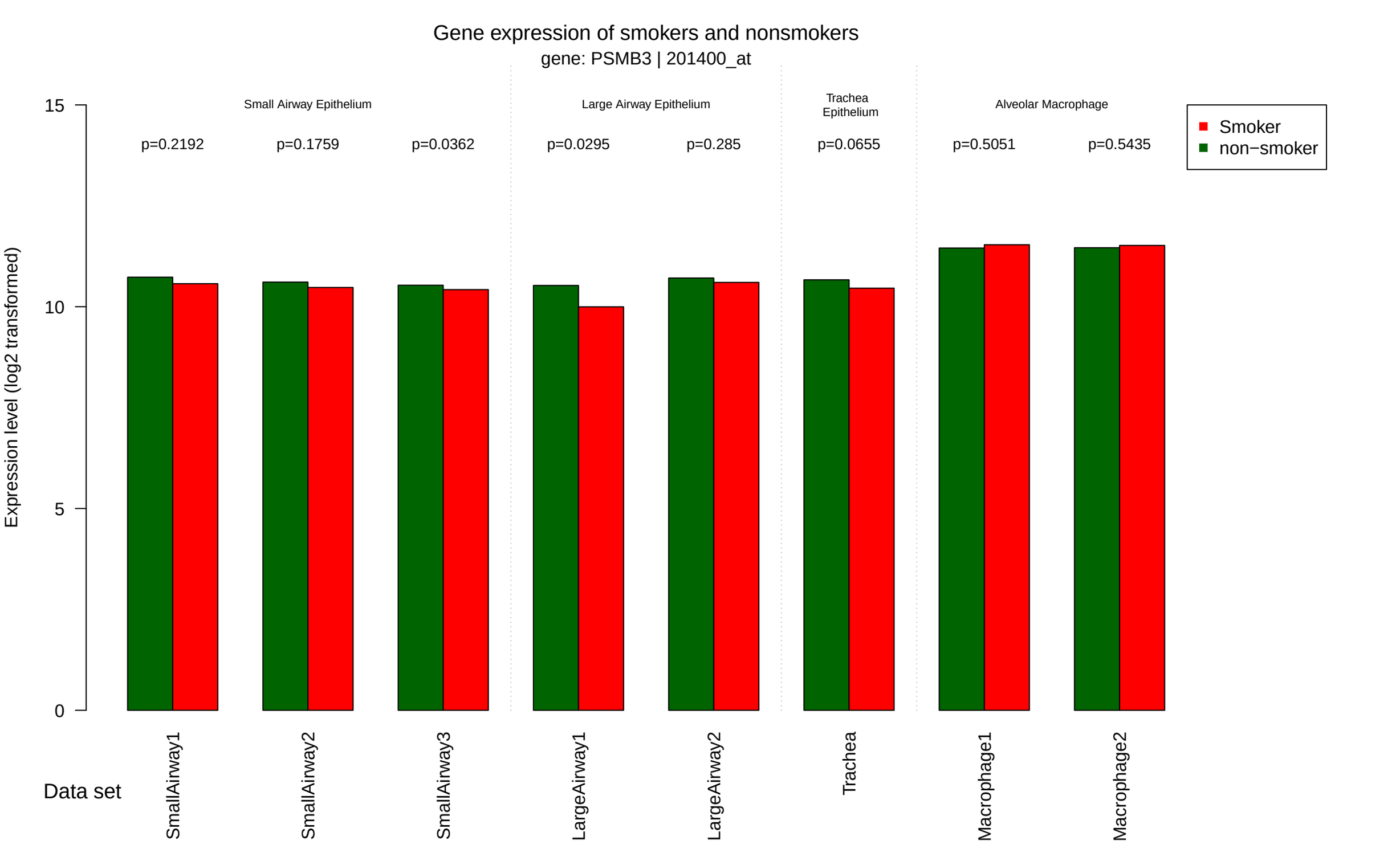
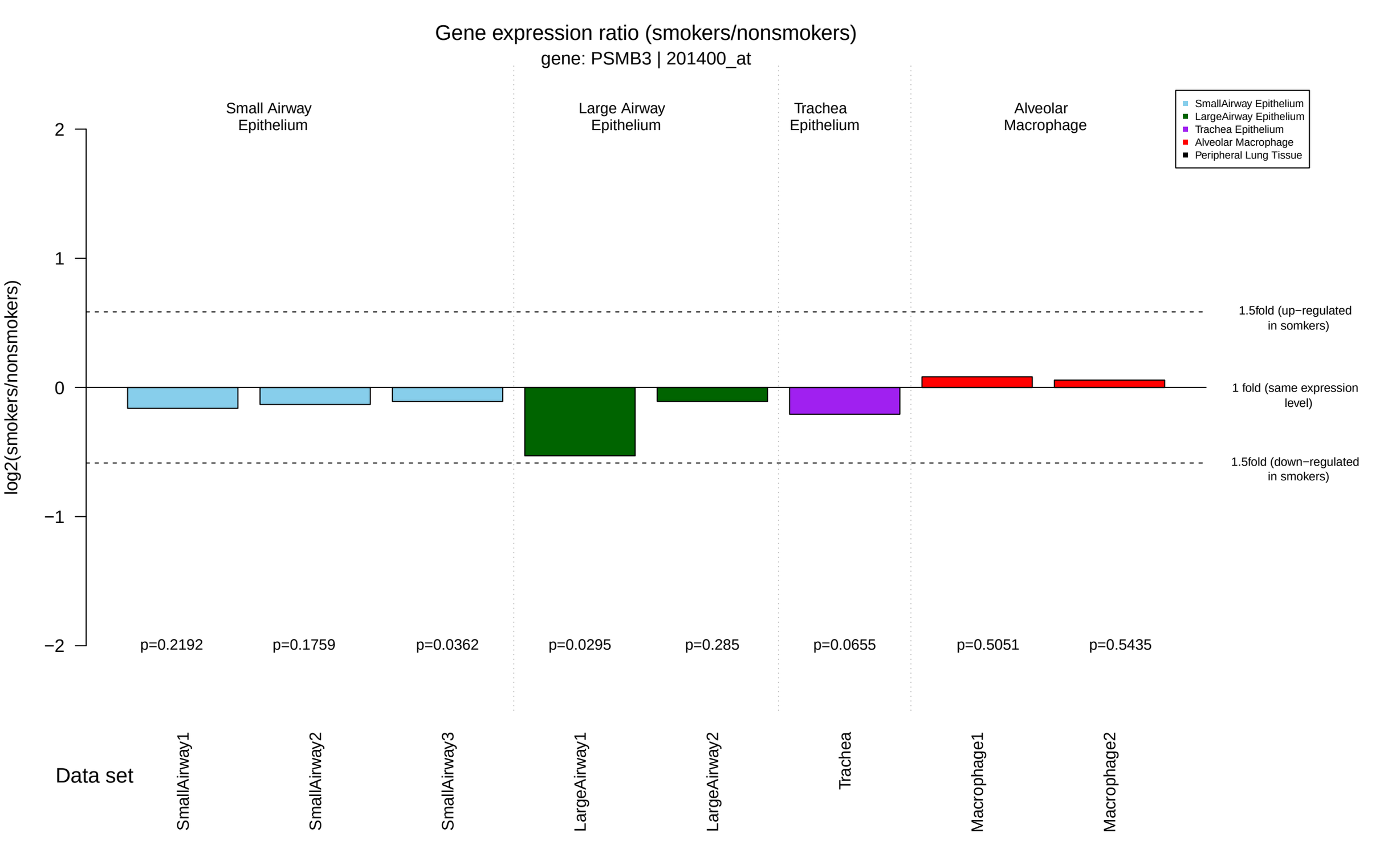
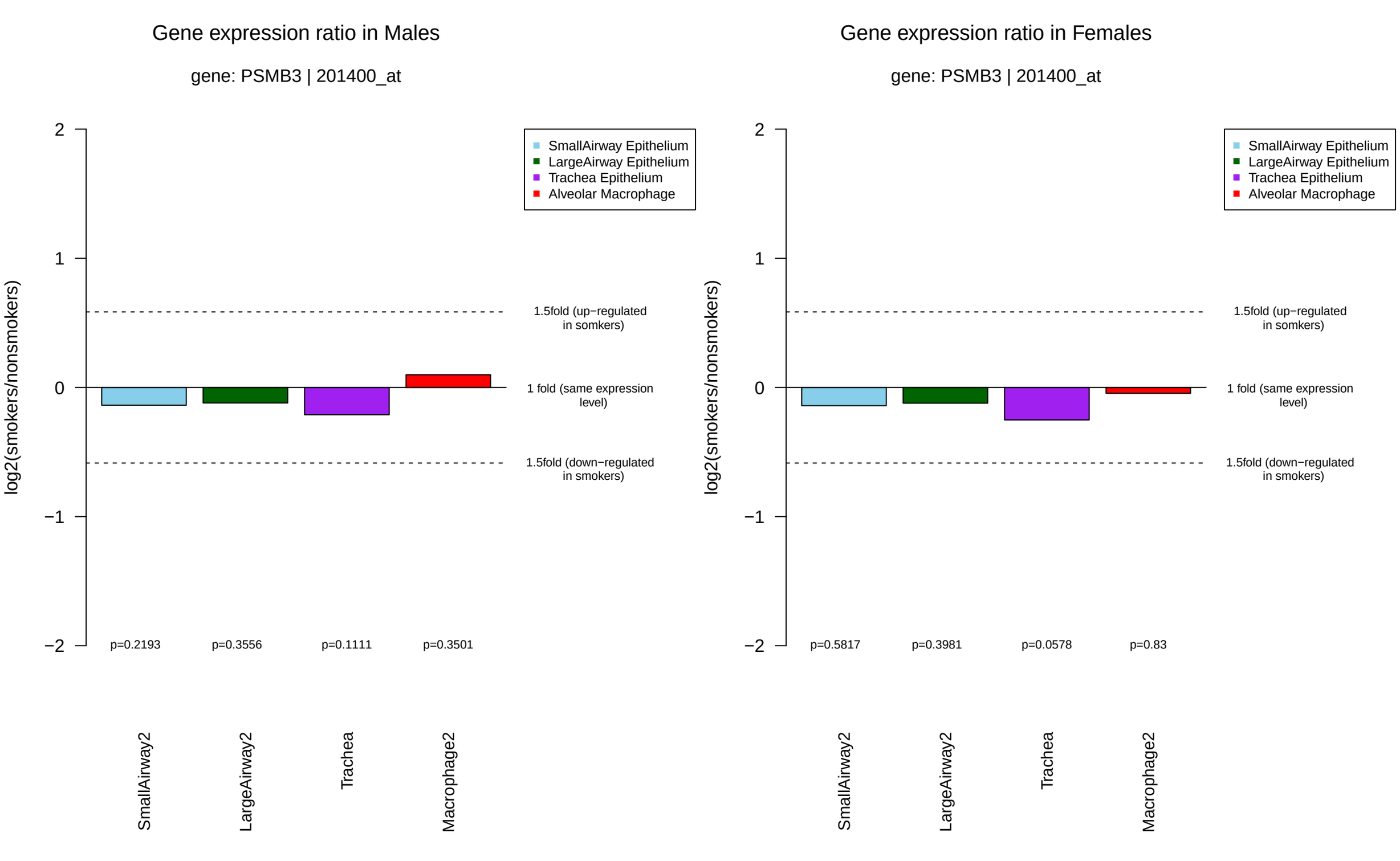
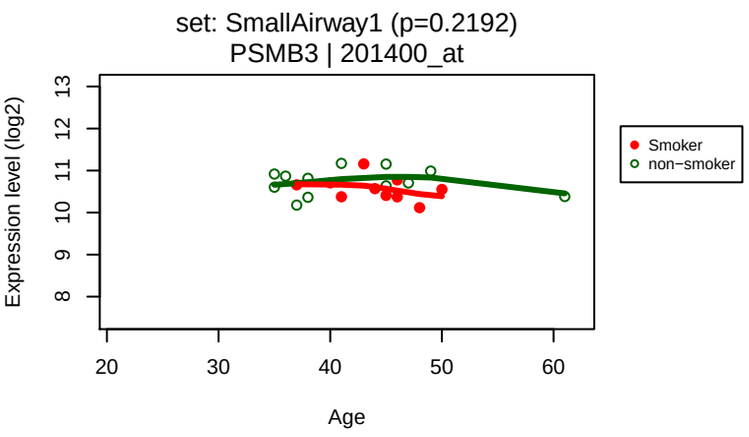
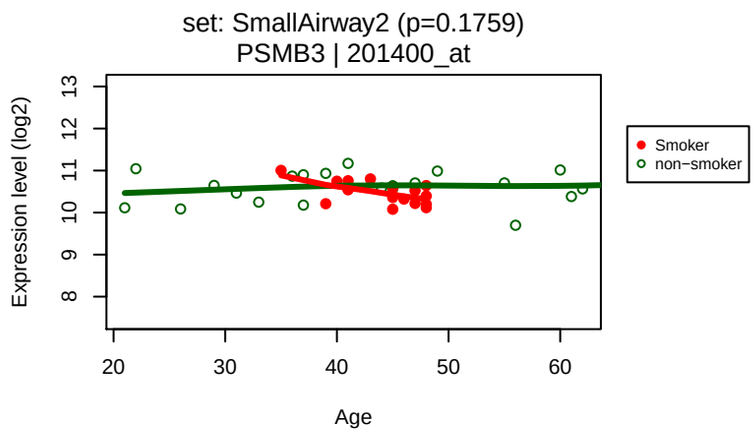
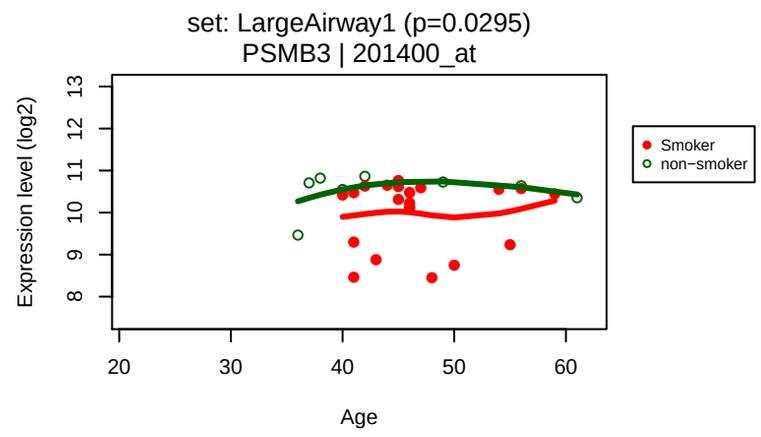

Gene expression of smokers and nonsmokers
gene: PSMB3 | 201400_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2192
p=0.1759
p=0.0362
p=0.0295
p=0.285
p=0.0655
p=0.5051
p=0.5435
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

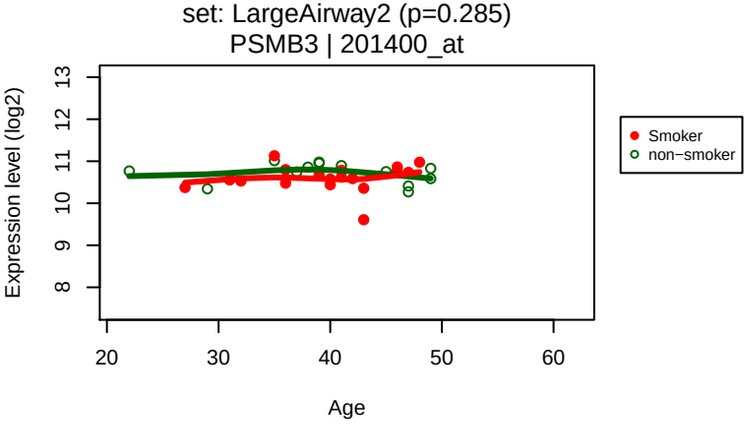

Gene expression ratio (smokers/nonsmokers)
gene: PSMB3 | 201400_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2192
p=0.1759
p=0.0362
p=0.0295
p=0.285
p=0.0655
p=0.5051
p=0.5435
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

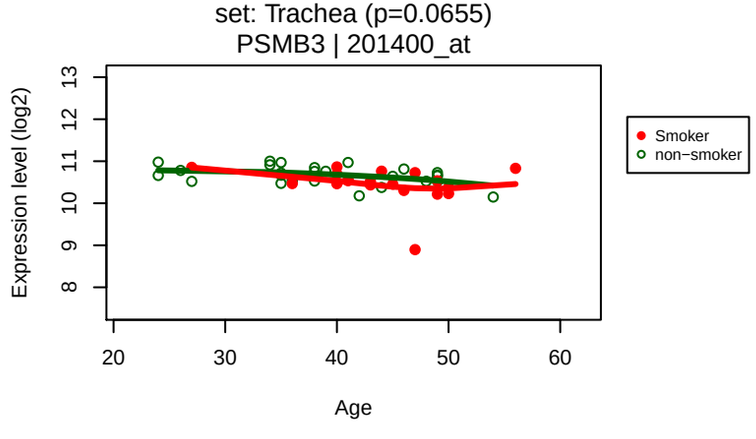

Gene expression ratio in Males
gene: PSMB3 | 201400_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2193
p=0.3556
p=0.1111
p=0.3501
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PSMB3 | 201400_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5817
p=0.3981
p=0.0578
p=0.83
SmallAirway2
LargeAirway2
Trachea
Macrophage2

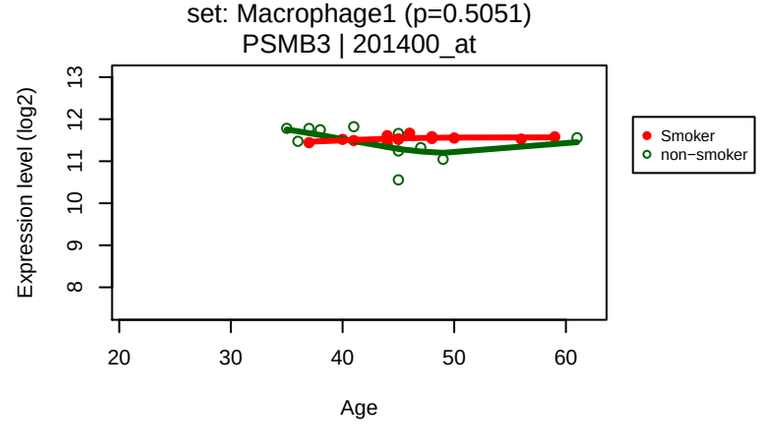

set: SmallAirway1 (p=0.2192)
PSMB3 | 201400_at
Expression level (log2)
Age
Smoker
non−smoker

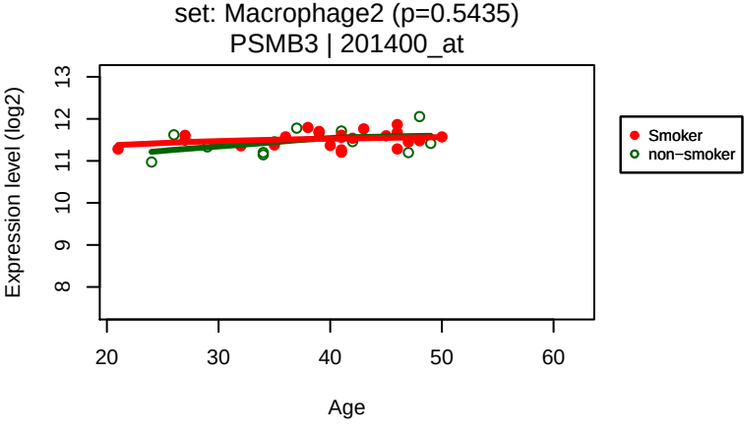

set: SmallAirway2 (p=0.1759)
PSMB3 | 201400_at
Expression level (log2)
Age
Smoker
non−smoker

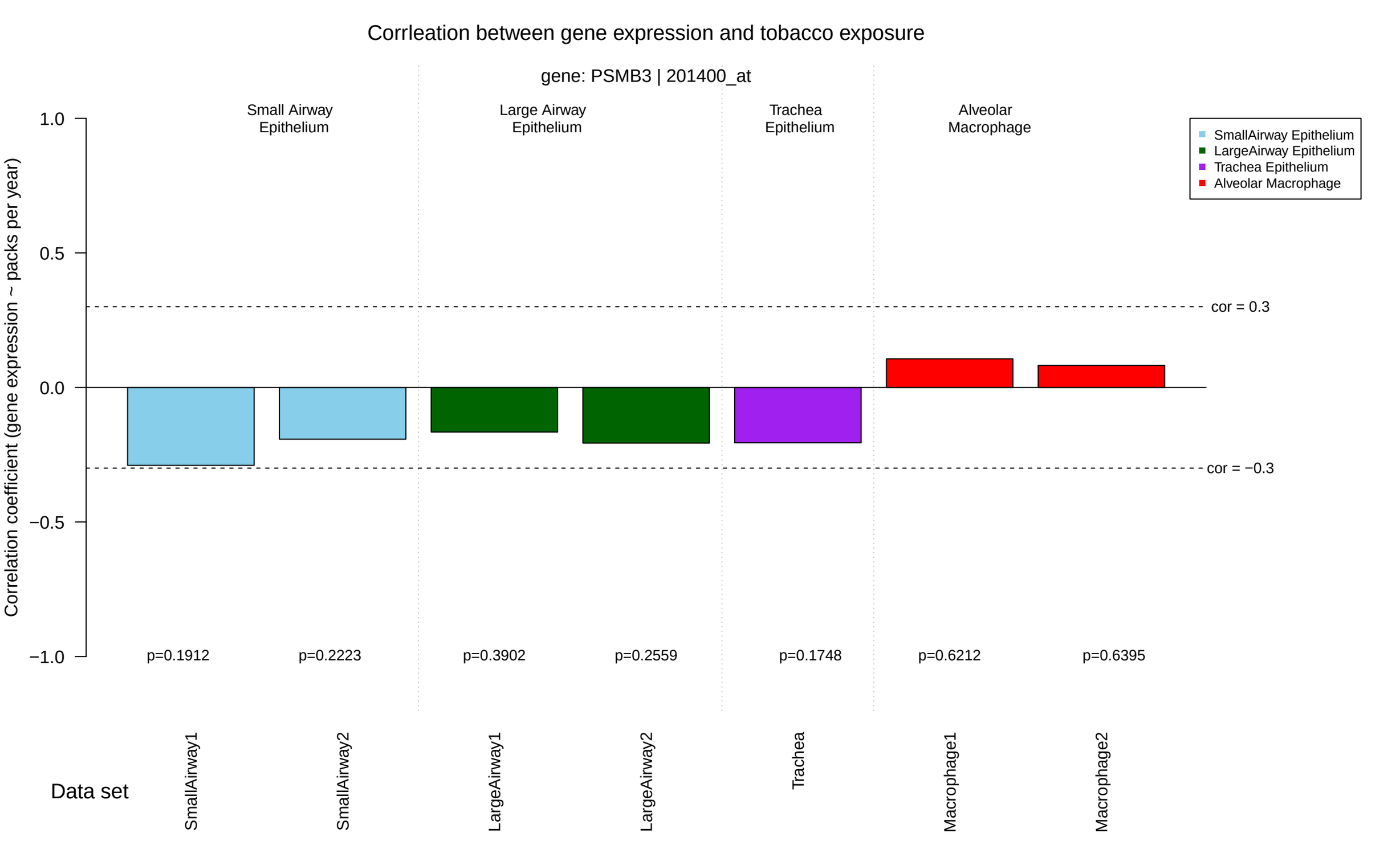

set: LargeAirway1 (p=0.0295)
PSMB3 | 201400_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.285)
PSMB3 | 201400_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0655)
PSMB3 | 201400_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5051)
PSMB3 | 201400_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5435)
PSMB3 | 201400_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PSMB3 | 201400_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1912
p=0.2223
p=0.3902
p=0.2559
p=0.1748
p=0.6212
p=0.6395
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2